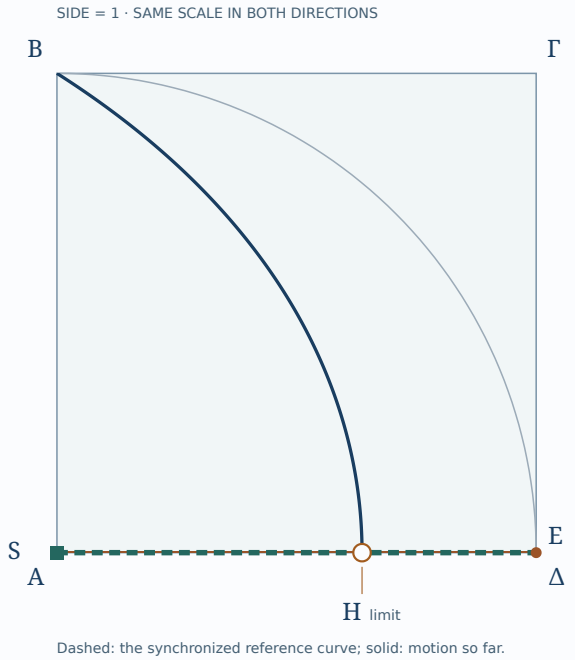
The point the motion cannot draw.

Selected clauses from Pappus, Collection IV.30–32, Hultsch 1876 I pp.250–259. Greek and Hultsch's Latin are retained edition readings; English, coordinates, clock experiments and limit calculation are new. This is not a complete translation or a physical instrument reconstruction. The program uses π to prescribe the motion; it does not construct or discover π independently.



Teal: translating line. Rust: rotating line. Blue: traced crossings. An open H, if shown, is a separately added limit. Dashed curve: synchronized reference.

Applied experiment

Time 1000000/1000000. Timing: together. Limiting point shown separately.

Slider height

0/1000000 = 0

Remaining turn

0/1000000 of a right angle

Intersection Z

Not unique · coincident lines

The lines coincide. Every point of the baseline belongs to both lines; their intersection does not select H. No traced point Z exists at this instant.

Approach, then reach

Synchronized motion · unit side · decimal approximations

Time left u	Crossing x	2/π − x	Line relation
0.5	0.5	0.13661977	One crossing
0.1	0.63137515	0.00524462	One crossing
0.01	0.63656741	0.00005236	One crossing
1.000000e-6	0.63661977	5.235988e-13	One crossing
Exactly 0	No unique x	Not a crossing	Coincident
Limit, added separately	2/π ≈ 0.63661977	0	Continuous extension

The last instant is not a crossing

Put A at (0, 0), B at (0, 1). With a fraction u of the synchronized journey left, the slider is at y = u and the rotating line makes θ = πu/2 with the baseline.

For u > 0, their unique crossing has
x = u cot(πu/2), y = u.

At u = 0, substituting into this expression gives an undefined product. The line equations give y = 0 twice. They do not give x = 2/π.

A separate argument adds H

For 0 < θ < π/2, the unit-circle triangle and sector comparison gives sin θ < θ < tan θ. Hence

cos θ < θ cot θ < 1.

Both bounds tend to 1 as θ tends to zero. Since x = (2/π) θ cot θ, the positive-time crossings approach **H = (2/π, 0)**.

This is a modern limit argument. It justifies a continuous extension of the curve; it does not turn the final coincident lines into one crossing. Pappus's geometric proof is retained on pp.256–259.

The program supplies π to draw the motion. A numerical animation cannot establish that an exact physical mechanism or a straightedge-and-compass construction has been supplied.

Why this point matters

From intercept to quarter-circle

Keep the unit side AB = 1. If AH = 2/π is supplied by the extended curve, the proportion is

$$AH : AB = AB : L,$$
$$L = 1/AH = \pi/2.$$

L is the straight length equal to the quarter-circle arc. Four copies give C = 2π. This step consumes the intercept; it does not obtain it by a straightedge and compass.

From circumference to equal area

A unit circle has the area of a right triangle with base C and height 1. The equivalent rectangle has sides C/2 and 1; a mean proportional gives square side √(C/2) = √π.

Pappus invokes Archimedes for the area step on pp.258–259. Euclid II.14 explains the rectangle-to-square construction. A numerical √π computed by this program is not a straightedge-and-compass squaring of the circle.

Read the generation and the objection

1. A name from a property

GREEK · P.252

καλεῖται γὰρ ὑπ’ αὐτῶν τετραγωνίζουσα καὶ γένεσιν ἔχει τοιαύτην.

HULTSCH’S LATIN · P.253

Quadratrix enim ab illis vocatur et huiusmodi ortum habet.

For they call it the squaring curve, and it comes into being in the following way.

The preceding sentence names Dinostratus, Nicomedes and other later geometers as users of the curve. This passage does not supply a biography of its inventor. The participle names a task: making square.

τετραγωνίζουσα / quadratrix

‘squaring’: a feminine participle agreeing with the understood γραμμή, line or curve.

γένεσιν / ortum

coming into being, generation. Here the following motion describes how the curve is produced.

---

2. In the same time

GREEK · P.252

καὶ ἐν ἴσῳ χρόνῳ ἥ τε ΑΒ κινουμένη ὁμαλῶς τὴν ὑπὸ ΒΑΔ γωνίαν, τουτέστιν τὸ Β σημεῖον τὴν ΒΕΔ περιφέρειαν, διανυέτω, καὶ ἡ ΒΓ τὴν ΒΑ εὐθεῖαν παροδευέτω, τουτέστιν τὸ Β σημεῖον κατὰ τῆς ΒΑ φερέσθω.

HULTSCH’S LATIN · P.253

atque eodem tempore et recta AB aequabiliter progrediens angulum BAΔ, id est punctum B circumferentiam BEΔ, percurrat, et recta BΓ ipsam BA praetervehatur, id est punctum B rectam BA permeet.

In the same time, let AB move uniformly through the angle BAD—that is, let the point B traverse the arc BED—and let BG travel over the straight line BA, that is, let the point B move along BA.

The source uses B for two moving points: the end rotating along the arc and the end sliding down BA. Hultsch explains this in his p.253 footnote. Our plate calls them E and S to keep them distinct. Both motions are uniform in this model; reading 4 supplies their proportional progress.

ἐν ἴσῳ χρόνῳ / eodem tempore

in an equal time. Equal arrival times do not mean equal distances or equal linear speeds.

ὁμαλῶς / aequabiliter

uniformly, at an even rate. This is a condition on the motion, not an effect of drawing a smooth curve.

3. One moving intersection

GREEK · P.252

τοιαύτης δὴ γινομένης κινήσεως τεμοῦσιν ἀλλήλας ἐν τῇ φορᾷ αἱ ΒΓ ΒΑ εὐθεῖαι κατά τι σημεῖον αἰεὶ συμμεθιστάμενον αὐταῖς

HULTSCH'S LATIN · P.253

Itaque, dum huiusmodi motus peragitur, rectae BA BΓ in eo motu sese invicem secabunt in puncto aliquo semper cum ipsis progrediente

As this motion takes place, the straight lines BG and BA intersect one another during their movement at a point that continually changes position along with them.

This is a selected clause, ending before the description of the curve it draws. Follow Z: it must lie on both moving lines, not merely on a pre-drawn curve. In the timing experiments the full lines can meet outside the square, or cease to meet altogether.

τεμοῦσιν ἀλλήλας / sese invicem secabunt

they will cut one another. The future tense describes the consequence of the prescribed motion.

συμμεθιστάμενον / cum ipsis progrediente

changing position together with them; a participle describing the intersection point.

---

4. The same fraction of the journey

GREEK · P.252

ἥτις γὰρ ἂν διαχθῇ τυχοῦσα πρὸς τὴν περιφέρειαν, ὡς ἡ ΑΖΕ, ἔσται ὡς ὅλη ἡ περιφέρεια πρὸς τὴν ΕΔ, ἡ ΒΑ εὐθεῖα πρὸς τὴν ΖΘ· τοῦτο γὰρ ἐκ τῆς γενέσεως τῆς γραμμῆς φανερόν ἐστιν.

HULTSCH'S LATIN · P.253

Principalis autem eius proprietas haec est, ut, si quaelibet recta, velut AZE, ad circumferentiam ducatur, sit ut tota circumferentia BEΔ ad EΔ, ita recta BA ad ZΘ; hoc enim ex ortu lineae manifestum est.

For any line drawn to the circumference, such as AZE, the whole arc is to ED as the straight line BA is to ZΘ. This follows plainly from the generation of the curve.

With the side set to one, this says remaining arc / quarter-arc = ZΘ / BA = u. The experiment writes the angle in radians as θ = πu/2. That coordinate notation and the function cotangent are modern explanatory tools.

περιφέρεια / circumferentia

circumference or arc, depending on context. Here the ‘whole’ arc is only a quarter-circle.

ὡς … πρὸς … / ut … ad …

as … is to …: the two matching comparisons state a proportion.

---

5. A ratio already assumed?

GREEK · P.254

ἐν γὰρ τούτῳ τῷ λόγῳ καὶ τὰ τάχη τῶν κινήσεων ἀνάγκη εἶναι.

HULTSCH’S LATIN · P.255

Nam in eadem proportione etiam celeritates motuum esse oportet.

For the speeds of the motions must also be in this ratio.

Pappus reports Sporus’s question: how can the two points arrive together without first knowing the ratio of the straight path to the arc? These are distances travelled per time, not an equality between angular and linear units. This is a reported objection; Hultsch’s note on p.253 disputes its force. The program supplies π when programming the motion.

λόγῳ / proportione

ratio, in this geometric setting. The same Greek word can mean an account or argument elsewhere.

τάχη / celeritates

speeds, plural. Matching completion times for different path lengths requires different speeds.

---

6. The missing end

GREEK · P.254

ἔπειτα δὲ τὸ πέρας αὐτῆς ᾧ χρῶνται πρὸς τὸν τετραγωνισμὸν τοῦ κύκλου, τουτέστιν καθ’ ὃ τέμνει σημεῖον τὴν ΑΔ εὐθεῖαν, οὐχ εὑρίσκεται.

HULTSCH’S LATIN · P.255

Praeterea lineae terminus, quem ad quadraturam circuli adhibent, id est punctum, in quo linea rectam AΔ secat, nequaquam invenitur.

Next, its endpoint, which they use for squaring the circle—that is, the point where it cuts the straight line AD—is not found.

The negative concerns obtaining the point by the stated generation. It is not a claim that the mathematical limit does not exist. Stop a little before the finish, then reach the finish exactly.

πέρας / terminus

end or boundary. Here it is the intercept used in the quadrature, not the current moving point.

οὐχ εὑρίσκεται / nequaquam invenitur

is not found. Do not turn this statement about the procedure into ‘there is no limit’.

---

7. Coincidence removes the unique point

GREEK · P.254

ὁπόταν γὰρ αἱ ΓΒ ΒΑ φερόμεναι συναποκατασταθῶσιν, ἐφαρμόσουσιν τῇ ΑΔ καὶ τομὴν οὐκέτι ποιήσουσιν ἐν ἀλλήλαις

HULTSCH’S LATIN · P.255

Nam si rectae AB BΓ simul ad finem motus deductae erunt, cum ipsa AΔ congruent neque amplius sese invicem secabunt.

When the moving lines GB and BA arrive together, they coincide with AD and no longer make an intersection with one another.

The Greek selection ends before its explanatory continuation. In modern set language the coincident lines have infinitely many common points. ‘No intersection’ here means no distinct crossing that selects the moving point. Our end state therefore records ‘coincident’, with no unique Z; it never substitutes H for that result.

ἐφαρμόσουσιν / congruent

they will fit or coincide. This is different from crossing at one point.

οὐκέτι / neque amplius

no longer: the change at the exact final instant matters.

---

8. The theorem after the objection

GREEK · P.256

δείκνυται, ὡς ἡ ΔΕΒ περιφέρεια πρὸς τὴν ΒΓ εὐθεῖαν, οὕτως ἡ ΒΓ πρὸς τὴν ΓΘ εὐθεῖαν.

HULTSCH’S LATIN · P.257

demonstratur esse ut circumferentiam BEΔ ad rectam BΓ, ita rectam BΓ ad ΓΘ.

It is proved that the arc DEB is to the straight line BG as BG is to the straight line GΘ.

The source changes its diagram labels: Γ is now the centre and Θ the intercept, corresponding to A and H in our motion plate. For a unit side, the theorem is (π/2) : 1 = 1 : AH. Pappus gives a geometric argument by contradiction on pp.256–259. Our limit calculation is a different, modern argument. The following paragraph explains rectification and then squaring, invoking Archimedes.

δείκνυται / demonstratur

it is shown or proved. The text continues with mathematical reasoning after reporting the objections.

εὐθεῖαν / rectam

straight line, with γραμμή understood. In a ratio this means the length of the indicated segment.

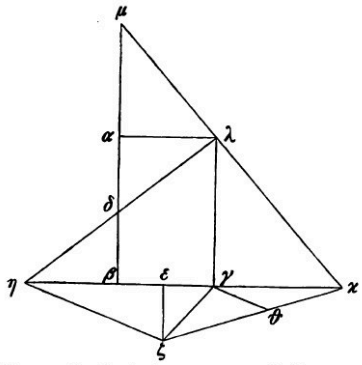
The complete printed pages

Pappus, edited and translated into Latin by Friedrich Hultsch, 1876, volume I, printed pp.250–259. PDF pages 283–292. Google scan of Oxford University's copy; public domain. Complete pages reduced to 1800 pixels on the long edge without cropping or retouching. Greek theta and geometric labels are normalized in the selected transcription; the scans retain the print and apparatus.

The later proof changes the diagram's centre from A to Γ and its intercept from H to Θ. Hultsch's footnote on p.253 disputes the force of the objections. The p.254–256 apparatus preserves editorial intervention; this study is not a new critical edition.

Complete source PDF

οὕτως ἡ ΒΓ πρὸς ΓΚ. καὶ ἔστι τῆς μὲν ΑΒ ἡμίσεια ἡ ΑΔ, τῆς δὲ ΒΓ διπλῆ ἡ ΓΗ· ἔσται ἄρα καὶ ὡς ἡ ΜΑ πρὸς ΑΔ, οὕτως ἡ ΗΓ πρὸς ΚΓ. ἀλλ' ὡς ἡ ΗΓ πρὸς ΓΚ, οὕτως ἡ ΖΘ πρὸς ΘΚ διὰ τὰς παραλλήλους τὰς ΗΖ ΓΘ· καὶ συνθέντι ἄρα ὡς ἡ ΜΔ πρὸς ΔΑ, ἡ ΖΚ πρὸς ΚΘ. ἴση δὲ ὑπόκειται καὶ ἡ ΑΔ τῇ ΘΚ [ἐπεὶ καὶ τῇ ΓΖ ἴση ἐστὶν ἡ ΑΔ]· ἴση ἄρα καὶ ἡ ΜΔ τῇ ΖΚ· ἴσον ἄρα καὶ τὸ ἀπὸ ΜΔ τῷ ἀπὸ ΖΚ. καὶ ἔστι τῷ μὲν ἀπὸ ΜΔ ἴσον τὸ ὑπὸ ΒΜΑ μετὰ τοῦ ἀπὸ ΔΑ, τῷ δὲ ἀπὸ ΖΚ ἴσον ἐδείχθη τὸ ὑπὸ ΒΚΓ μετὰ τοῦ ἀπὸ ΖΓ, ὧν τὸ ἀπὸ ΑΔ ἴσον τῷ ἀπὸ ΓΖ (ἴση γὰρ ὑπόκειται ἡ ΑΔ τῇ ΓΖ)· ἴσον ἄρα καὶ τὸ ὑπὸ ΒΜΑ τῷ ὑπὸ ΒΚΓ· ὡς ἄρα ἡ ΜΒ πρὸς ΒΚ, ἡ ΓΚ πρὸς ΜΑ. ἀλλ' ὡς ἡ ΒΜ πρὸς ΒΚ, ἡ ΑΓ πρὸς ΓΚ· ὡς ἄρα ἡ ΑΓ πρὸς ΓΚ, ἡ ΓΚ πρὸς ΑΜ. ἔστι δὲ καὶ ὡς ἡ ΜΒ πρὸς ΒΚ, ἡ ΜΑ πρὸς ΑΔ· καὶ ὡς ἄρα ἡ ΑΓ πρὸς ΓΚ, ἡ ΓΚ πρὸς ΑΜ, καὶ ἡ ΑΜ πρὸς ΑΔ.



44 κϑ'. *Τούτου δειχθέντος πρόδηλον ὅπως δεῖ κύβου δοθέντος κύβον ἄλλον εὑρεῖν κατὰ τὸν δοθέντα λόγον.*

Ἔστω γὰρ ὁ δοθεὶς λόγος τῆς Α εὐθείας πρὸς τὴν Β, καὶ τῶν Α Β δύο μέσαι ἀνάλογον κατὰ τὸ συνεχὲς εἰλήφθωσαν αἱ Γ Δ· ἔσται ἄρα ὡς ἡ Α πρὸς τὴν Β, οὕτως ὁ ἀπὸ τῆς Α κύβος πρὸς τὸν ἀπὸ τῆς Γ κύβον· τοῦτο γὰρ δῆλον ἐκ τῶν στοιχείων.

45 λ'. *Εἰς τὸν τετραγωνισμὸν τοῦ κύκλου παρελήφθη τις*

2. post ἡ ΓΗ add. ἐπεὶ καὶ ἡ ΑΓ τῇ ΑΔ Eutoc. 8. τὰς ΗΖ ΓΘ Pappus p. 62, 4 Eutoc., ἡ $\overline{ΖΓΘ}$ AB^1, $\overline{ηζ}$ $\overline{γϑ}$ B^{3}S 12. ἐπεὶ — 14. ἡ ΑΔ] conf. supra ad p. 62, 2. 3 17. τὸ ὑπὸ ΒΜΑ Pappus p. 62, 5

$\mu\alpha : \alpha\beta = \beta\gamma : \gamma\varkappa$. Et est $\alpha\beta = 2\,\alpha\delta$, et $\beta\gamma = \frac{1}{2}\eta\gamma$ (*quia* $\eta\beta = \alpha\lambda = \beta\gamma$); ergo erit etiam

$\mu\alpha : \alpha\delta = \eta\gamma : \gamma\varkappa$. Sed propter parallelas $\eta\zeta$ $\gamma\vartheta$ est $\eta\gamma : \gamma\varkappa = \zeta\vartheta : \vartheta\varkappa$; ergo etiam componendo

$\mu\delta : \alpha\delta = \zeta\varkappa : \vartheta\varkappa$. Sed ex hypothesi est $\alpha\delta = \vartheta\varkappa$; ergo etiam $\mu\delta = \zeta\varkappa$, et

$\mu\delta^2 = \varkappa\zeta^2$. Et *propter elem. 2, 6* est

$\mu\delta^2 = \beta\mu \cdot \mu\alpha + \alpha\delta^2$, et *supra* demonstratum est

$\varkappa\zeta^2 = \beta\varkappa \cdot \varkappa\gamma + \gamma\zeta^2$. Iam in his est $\alpha\delta^2 = \gamma\zeta^2$ (nam ex hypothesi est $\alpha\delta = \gamma\zeta$); ergo etiam *subtrahendo*

$\beta\mu \cdot \mu\alpha = \beta\varkappa \cdot \varkappa\gamma$, id est *proportione facta* (*elem. 6, 16*)

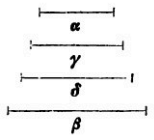
$\mu\beta : \beta\varkappa = \gamma\varkappa : \mu\alpha$. Sed *propter parallelas* $\mu\beta$ $\lambda\gamma$ est $\mu\beta : \beta\varkappa = \lambda\gamma : \gamma\varkappa$; ergo etiam

$\lambda\gamma : \gamma\varkappa = \gamma\varkappa : \mu\alpha$. Sed *propter parallelas* $\beta\varkappa$ $\alpha\lambda$ est $\mu\beta : \beta\varkappa = \mu\alpha : \alpha\lambda$, *ideoque*

$\gamma\varkappa : \mu\alpha = \mu\alpha : \alpha\lambda$; ergo etiam

$\lambda\gamma : \gamma\varkappa = \gamma\varkappa : \mu\alpha = \mu\alpha : \alpha\lambda$.

XXIX. Hoc demonstrato apparet, quomodo, dato cubo, alium cubum iuxta datam proportionem invenire oporteat. Prop. 25



Sit enim data proportio $\alpha : \beta$, et rectarum $\alpha\,\beta$ duae mediae in continua proportione sumantur γ δ; erit igitur $\alpha : \beta = \alpha^3 : \gamma^3$; hoc enim manifestum est ex elementis (*5 def. 11. 8, 12. 11, 33*).

XXX. Ad circuli quadraturam a Dinostrato[1], Nicomede

1) Conf. Chasles, *Aperçu historique* etc., p. 28 versionis German., Bretschneider, *Geometrie vor Euklides*, p. 96, Herm. Hankel, *Geschichte der Mathematik*, p. 151.

Eutoc., τὸ ἀπὸ $\overline{BMA}$ ABS 21. ἡ ΓK πρὸς MA Pappus p. 62, 9. 10, ἡ $K\Gamma$ πρὸς AM Eutoc. V², ἡ $\overline{A\Gamma}$ πρὸς $\overline{\Gamma K}$ ABSV 22. καὶ ante ὡς ἄρα add. Pappus p. 62, 10 Eutoc. V² 22. 23. ἡ ΓK πρὸς AM — ἡ MB πρὸς BK Pappus p. 62, 11. 12, ἡ ΓK πρὸς AM. ἔστι δὲ καὶ ὡς ἡ $A\Gamma$ πρὸς ΓK Eutoc., om. ABS 26. $\overline{K\Theta}$ A¹ in marg. (BS) 29. τῶν $\overline{AB}$ AS, distinx. B 30. αἱ $\overline{\Gamma\Delta}$ A, distinx. BS 33. $\bar{\lambda}$ A¹ in marg. (BS)

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ὑπὸ Δεινοστράτου καὶ Νικομήδους γραμμὴ καί τινων ἄλλων νεωτέρων ἀπὸ τοῦ περὶ αὐτὴν συμπτώματος λαβοῦσα τοὔνομα· καλεῖται γὰρ ὑπ' αὐτῶν τετραγωνίζουσα καὶ γένεσιν ἔχει τοιαύτην.

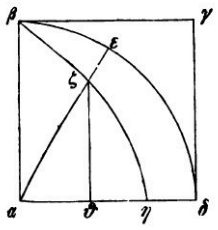
Ἐκκείσθω τετράγωνον τὸ ΑΒΓΔ καὶ περὶ κέντρον τὸ Α περιφέρεια γεγράφθω ἡ ΒΕΔ, καὶ κινείσθω ἡ μὲν ΑΒ οὕτως ὥστε τὸ μὲν Α σημεῖον μένειν τὸ δὲ Β φέρεσθαι κατὰ τὴν ΒΕΔ περιφέρειαν, ἡ δὲ ΒΓ παράλληλος ἀεὶ διαμένουσα τῇ ΑΔ τῷ Β σημείῳ φερομένῳ κατὰ τῆς ΒΑ συνακολουθείτω, καὶ ἐν ἴσῳ χρόνῳ ἥ τε ΑΒ κινουμένη ὁμαλῶς τὴν ὑπὸ ΒΑΔ γωνίαν, τουτέστιν τὸ Β σημεῖον τὴν ΒΕΔ περιφέρειαν, διανύέτω, καὶ ἡ ΒΓ τὴν ΒΑ εὐθεῖαν παροδευέτω, τουτέστιν τὸ Β σημεῖον κατὰ τῆς ΒΑ φερέσθω. συμβήσεται δῆλον τῇ ΑΔ εὐθείᾳ ἅμα ἐφαρμόζειν ἑκατέραν τήν τε ΑΒ καὶ τὴν ΒΓ. τοιαύτης δὴ γινομένης κινήσεως τεμοῦσιν ἀλλήλας ἐν τῇ φορᾷ αἱ ΒΓ ΒΑ εὐθεῖαι κατά τι σημεῖον αἰεὶ συμμεθιστάμενον αὐταῖς, ὑφ' οὗ σημείου γράφεταί τις ἐν τῷ μεταξὺ τόπῳ τῶν τε ΒΑΔ εὐθειῶν καὶ τῆς ΒΕΔ περιφερείας γραμμὴ ἐπὶ τὰ αὐτὰ κοίλη, οἷα ἐστὶν ἡ ΒΖΗ, ἣ καὶ χρειώδης εἶναι δοκεῖ πρὸς τὸ τῷ δοθέντι κύκλῳ τετράγωνον ἴσον εὑρεῖν. τὸ δὲ ἀρχικὸν αὐτῆς σύμπτωμα τοιοῦτόν ἐστιν. ἥτις γὰρ ἂν διαχθῇ τυχοῦσα πρὸς τὴν περιφέρειαν, ὡς ἡ ΑΖΕ, ἔσται ὡς ὅλη ἡ περιφέρεια πρὸς τὴν ΕΔ, ἡ ΒΑ εὐθεῖα πρὸς τὴν ΖΘ· τοῦτο γὰρ ἐκ τῆς γενέσεως τῆς γραμμῆς φανερόν ἐστιν.

46 λα'. Δυσαρεστεῖται δὲ αὐτῇ ὁ Σπόρος εὐλόγως διὰ

1. ὑπὸ *νικοστράτου* B *νικοδήμου* AB^3, *νικομήδου* B^1 *To*, corr. S 5. τὸ $\overline{AB}$ $\overline{\Gamma\Delta}$ AS, coniunx. B καὶ add. *To* auctore *Co* 6. ἡ add. *Hu* 9. τῶι $\overline{B}$ A *To*, τὸ $\bar{\beta}$ B, τῷ et τὸ $\bar{\beta}$ Paris. 2368 S σημεῖον φέρον ἐν ὧι ABS, corr. *To* κατὰ τῆς *BA To* pro κατὰ τῆς $\overline{B}$ 10. συνακολουθεῖ τῶι AS, συνακολουθεῖ τὸ B, corr. *Sca To* καὶ add. *To* κινουμένης AB^{3}S, corr. B^1 12. περιφέρειαν add. *To* auctore *Co* 13. παροδευέτω *Hu*, ////ευέτω A (sed initio vestigia litterarum παρ comparent), ...ενέτω B^{1}S, παραλευέτω B^3, παραβεβέτω e "cod. Vat." affert atque inde παραβαινέτω scribit *To* 14. δῆλον] δὴ vel δηλονότι coni. *Hu* ἑκατερα A, ἑκατέρα B, corr. S 20. ἣ add. *Hu* χρειῶδες ABS, corr. *To* auctore *Co* 22. 23. πρὸς τὴν —

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aliisque nonnullis recentioribus linea quaedam adsumpta est, quae ex ipsius proprietate nomen accepit. Quadratix enim ab illis vocatur et huiusmodi ortum habet.



Exponatur quadratum $\alpha\beta\gamma\delta$, et circa centrum α circumferentia $\beta\varepsilon\delta$ describatur, et recta quidem $\alpha\beta$ ita moveatur, ut punctum α maneat ac β feratur in circumferentia $\beta\varepsilon\delta$, recta autem $\beta\gamma$, parallela semper ipsi $\alpha\delta$ manens, punctum β, dum fertur in $\beta\alpha$*), comitetur, atque eodem tempore et recta $\alpha\beta$ aequabiliter progrediens angulum $\beta\alpha\delta$, id est punctum β circumferentiam $\beta\varepsilon\delta$, percurrat, et recta $\beta\gamma$ ipsam $\beta\alpha$ praetervehatur, id est punctum β rectam $\beta\alpha$ permeet. Eveniet igitur, ut simul et $\alpha\beta$ et $\beta\gamma$ cum recta $\alpha\delta$ congruant. Itaque, dum huiusmodi motus peragitur, rectae $\alpha\beta$ $\beta\gamma$ in eo motu se invicem secabunt in puncto aliquo semper cum ipsis progrediente, a quo puncto in spatio inter rectas $\beta\alpha$ $\alpha\delta$ et circumferentiam $\beta\varepsilon\delta$ linea quaedam ad easdem partes concava, qualis est $\beta\zeta\eta$, describitur, quae quidem utilis esse videtur ad quadratum dato circulo aequale inveniendum. Principalis autem eius proprietas haec est, ut, si quaelibet recta, velut $\alpha\zeta\varepsilon$, ad circumferentiam ducatur, sit ut tota circumferentia $\beta\varepsilon\delta$ ad $\varepsilon\delta$, ita recta $\beta\alpha$ ad $\zeta\vartheta$; hoc enim ex ortu lineae manifestum est.

XXXI. Sed ea linea Sporo his de causis iure displicet [1]).

*) Punctum β, pro utriusque rectarum $\alpha\beta$ $\beta\gamma$ peculiari motu, diverso sensu accipiendum esse apparet; est enim I: rectae $\alpha\beta$ terminus ac fertur in circuli quadrante, II: rectae $\beta\gamma$ initium ac fertur in $\beta\alpha$, totam $\beta\gamma$ parallelam sibi ipsi semper manentem secum trahens.

1) Haec argumenta, quae contra lineae constructionem afferuntur, non ita magni momenti esse demonstrat Bretschneider l. c.

ὅλη η add. *Hu* auctore *Co* (εὐθεῖα πρὸς τὴν περιφέρειαν, ὡς ἡ *BZE*, ἔσται ὅλη ἡ *BEΔ* add. *To*) 24. ἡ *BΔ*] ἡ $\overline{BA}$ περιφέρεια AB, ἡ *BΓ To*, corr. S *Co* 26. λα (sic) A^1 in marg. (S), om. B αὐτῷ (propter proximum λαμβάνει: conf. p. 255 adnot. 2) coni. *Hu*

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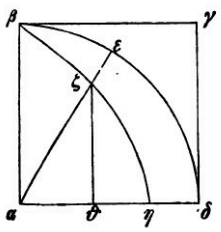
ταῦτα. πρῶτον μὲν γὰρ πρὸς ὃ δοκεῖ χρειώδης εἶναι πρᾶγμα, τοῦτ' ἐν ὑποθέσει λαμβάνει. πῶς γὰρ δυνατόν, δύο σημείων ἀρξαμένων ἀπὸ τοῦ Β κινεῖσθαι, τὸ μὲν κατ' εὐθείας ἐπὶ τὸ Α, τὸ δὲ κατὰ περιφερείας ἐπὶ τὸ Δ ἐν ἴσῳ χρόνῳ συναποκαταστῆσαι μὴ πρότερον τὸν λόγον τῆς ΑΒ εὐθείας πρὸς τὴν ΒΕΔ περιφέρειαν ἐπιστάμενον; ἐν γὰρ τούτῳ τῷ λόγῳ καὶ τὰ τάχη τῶν κινήσεων ἀνάγκη εἶναι. ἐπεὶ πῶς οἷόν τε συναποκαταστῆναι τάχεσιν ἀκρίτοις χρώμενα, πλὴν εἰ μὴ ἂν κατὰ τύχην ποτὲ συμβῇ; τοῦτο δὲ πῶς οὐκ ἄλογον; ἔπειτα δὲ τὸ πέρας αὐτῆς ᾧ χρῶνται πρὸς τὸν τετραγωνισμὸν τοῦ κύκλου, τουτέστιν καθ' ὃ τέμνει σημεῖον τὴν ΑΔ εὐθεῖαν, οὐχ εὑρίσκεται. νοείσθω δὲ ἐπὶ τῆς προκειμένης τὰ λεγόμενα καταγραφῆς· ὁπόταν γὰρ αἱ ΓΒ ΒΑ φερόμεναι συναποκατασταθῶσιν, ἐφαρμόσουσιν τῇ ΑΔ καὶ τομὴν οὐκέτι ποιήσουσιν ἐν ἀλλήλαις· παύεται γὰρ ἡ τομὴ πρὸ τῆς ἐπὶ τὴν ΑΔ ἐφαρμογῆς ἥπερ τομὴ πέρας αὖ ἐγένετο τῆς γραμμῆς, καθ' ὃ τῇ ΑΔ εὐθεία συνέπιπτεν. πλὴν εἰ μή λέγοι τις ἐπινοεῖσθαι προσεκβαλλομένην τὴν γραμμήν, ὡς ὑποτιθέμεθα τὰς εὐθείας, ἕως τῆς ΑΔ· τοῦτο δ' οὐχ ἕπεται ταῖς ὑποκειμέναις ἀρχαῖς, ἀλλ' ὡς ἂν ληφθείη τὸ Η σημεῖον προειλημμένου τοῦ τῆς περιφερείας πρὸς τὴν εὐθεῖαν λόγου. χωρὶς δὲ τοῦ δοθῆναι τὸν λόγον τοῦτον οὐ χρὴ τῇ τῶν εὑρόντων ἀνδρῶν δόξῃ πιστεύοντας παραδέχεσθαι τὴν γραμμὴν μηχανικωτέραν πως οὖσαν [καὶ

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2. post *δυνατὸν* add. *φησὶ To* (*inquit Co*) 4. *ἐπὶ τὸ* $\overline{\Delta E}$ $\overline{H}$ *ἴσωι* A, *ἐπὶ τὸ* $\overline{\delta\varepsilon\eta}$ *ἴσῳ* B, corr. S 5. *συναποκαταστῆναι* ABS, corr. *Hu* *τὸν λόγον* S, *τόλον* et initio superscr. *ὸν* A (prima manu), unde *τὸ ὅλον* B et Torelli "cod. Vat." 6. 7. *ἐν γὰρ τῷ αὐτῷ λόγῳ* coni. *Hu* 7. *ἀνάγκη εἶναι Hu*, *ἀναγκαῖον* ABS, *ἀναγκαῖον εἶναι To* (omisso posthac *ἐπεὶ*) 7. 8. *ἐπει πως οιονται* (sine acc.) A, *πῶς οἴονται γὰρ To*, corr. BS 8. *συναποκαταστῆσαι* coni. *Hu* *ἀκράτοις To* invitis ABS *χρώμενα* AB, *χρώμεθα* S, *χρώμενον* coni. *Hu* 9. *ἂν* del. *To* (probat ac *συμβαίη* coni. *Hu*) *ποτὲ To* pro *τότε* 13. *γὰρ* add. *Hu* 14. *τὴν* $\overline{\Delta\Delta}$ ABS, *ἐπὶ τὴν ΑΔ To*, corr. *Hu* 16. *πρὸ To* auctore *Co* pro *πρὸς* *αὖ Hu* pro *ἂν* 19. *ὑπο τε θέμεθα* A, corr. BS 20. *ἀλλ' ὡς δ' ἂν* AB *To*, *ἄλλως δ' ἂν* S, corr. *Hu* 21. 22. *προειλῆφθαι δεῖ τὸν — λόγον* voluit *Co* cum vertit: *sed utcumque sumatur*

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Primum enim, *inquit*, ad quod linea utilis esse videtur, id in hypothesi iste[2]) praesumit. Quomodo enim, si duo puncta a β moveri coeperint, efficias, ut alterum per rectam ad α, alterum per circumferentiam ad δ eodem tempore deducatur, nisi prius proportionem rectae $\alpha\beta$ ad circumferentiam cognoveris? Nam in eadem proportione etiam celeritates motuum esse oportet. Nempe quomodo *illa puncta*, celeritate motuum non definita, *eo quo diximus* deduci possunt, nisi aliquando casu fortuito id contingat? An vero id non absurdum est? Praeterea lineae terminus, quem ad quadraturam circuli adhibent, id est punctum, in quo linea rectam $\alpha\delta$ secat, nequaquam invenitur. Intellegantur autem haec quae dicimus in figura proposita. Nam si rectae $\alpha\beta$ $\beta\gamma$ simul *ad finem motus* deductae erunt, cum ipsa $\alpha\delta$ congruent neque amplius sese invicem secabunt. Desinunt enim secari, antequam cum $\alpha\delta$ congruunt, et tamen hanc ipsam sectionem *isti voluerunt* terminum lineae esse, in quo cum recta $\alpha\delta$ concurreret. Nisi forte quis dicat *ab extremo sectionis puncto* intellegi lineam usque ad $\alpha\delta$, *proinde* ac rectas *lineas* constituimus, produci; quod quidem ex iis quae initio supposita sunt minime sequitur, sed hoc *potius*, punctum η sumi non posse, nisi prius proportio circumferentiae ad rectam sumpta sit. Haec igitur proportio nisi data sit, cavendum est ne virorum qui *lineam* invenerunt auctoritati obsequentes *constructionem* eius admittamus, quippe quae ad mechanicam magis rationem accedat [et



2) Dinostratumne an Nicomedem an alium quempiam Sporus hoc loco dixerit, incertum est.

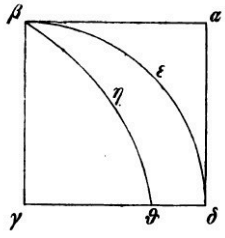
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punctum η, praecedere debet proportio circumferentiae ad rectam lineam, et similiter *To: ratio circumferentiae ad rectam utique praesumpta est,* id est *προείληπται ὁ — λόγος* 23. *οὐ Hu, η* A, *ἢ* BS, *ἀδύνατον. ἢ To* 24. *καὶ* — p. 256, 1. *μηχανικοῖς* interpolatori tribuit *Hu*

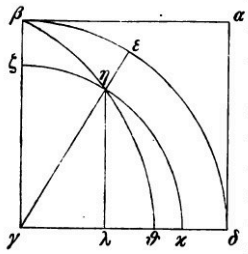
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εἰς πολλὰ προβλήματα χρησιμεύουσαν τοῖς μηχανικοῖς]. ἀλλὰ πρότερον παραδεκτέον ἐστὶ τὸ δι' αὐτῆς δεικνύμενον πρόβλημα.

47 Τετραγώνου γὰρ ὄντος τοῦ ΑΒΓΔ καὶ τῆς μὲν περὶ τὸ κέντρον τὸ Γ περιφερείας τῆς ΒΕΔ, τῆς δὲ ΒΗΘ τετραγωνιζούσης γινομένης, ὡς προείρηται, δείκνυται, ὡς ἡ ΔΕΒ περιφέρεια πρὸς τὴν ΒΓ εὐθεῖαν, οὕτως ἡ ΒΓ πρὸς τὴν ΓΘ εὐθεῖαν. εἰ γὰρ μὴ ἔστιν, ἤτοι πρὸς μείζονα ἔσται τῆς ΓΘ ἢ πρὸς ἐλάσσονα.



48 Ἔστω πρότερον, εἰ δυνατόν, πρὸς μείζονα τὴν ΓΚ, καὶ περὶ κέντρον τὸ Γ περιφέρεια ἡ ΖΗΚ γεγράφθω τέμνουσα τὴν γραμμὴν κατὰ τὸ Η, καὶ κάθετος ἡ ΗΛ, καὶ ἐπιζευχθεῖσα ἡ ΓΗ ἐκβεβλήσθω ἐπὶ τὸ Ε. ἐπεὶ οὖν ἐστιν ὡς ἡ ΔΕΒ περιφέρεια πρὸς τὴν ΒΓ εὐθεῖαν, οὕτως ἡ ΒΓ, τουτέστιν ἡ ΓΔ, πρὸς τὴν ΓΚ, ὡς δὲ ἡ ΓΔ πρὸς τὴν ΓΚ, ἡ ΒΕΔ περιφέρεια πρὸς τὴν ΖΗΚ περιφέρειαν (ὡς γὰρ ἡ διάμετρος τοῦ κύκλου πρὸς τὴν διάμετρον, ἡ περιφέρεια τοῦ κύκλου πρὸς τὴν περιφέρειαν), φανερὸν ὅτι ἴση ἐστὶν ἡ ΖΗΚ περιφέρεια τῇ ΒΓ εὐθείᾳ. καὶ ἐπειδὴ διὰ τὸ σύμπτωμα τῆς γραμμῆς ἐστιν ὡς ἡ ΒΕΔ περιφέρεια πρὸς τὴν ΕΔ, οὕτως ἡ ΒΓ πρὸς τὴν ΗΛ, καὶ ὡς ἄρα ἡ ΖΗΚ πρὸς τὴν ΗΚ περιφέρειαν, οὕτως ἡ ΒΓ εὐθεῖα πρὸς τὴν ΗΛ. καὶ ἐδείχθη ἴση ἡ ΖΗΚ περιφέρεια τῇ ΒΓ εὐθείᾳ· ἴση ἄρα καὶ ἡ ΗΚ περιφέρεια τῇ ΗΛ εὐθείᾳ, ὅπερ ἄτοπον. οὐκ ἄρα ἐστὶν ὡς ἡ ΒΕΔ περιφέρεια πρὸς τὴν ΒΓ εὐθεῖαν, οὕτως ἡ ΒΓ πρὸς μείζονα τῆς ΓΘ.



49 λβ'. Λέγω δὲ ὅτι οὐδὲ πρὸς ἐλάσσονα. εἰ γὰρ δυνατόν, ἔστω πρὸς τὴν ΚΓ, καὶ περὶ κέντρον τὸ Γ περιφέρεια

ad multa problemata sit mechanicis utilis]. Sed primum illud problema quod per lineam *quadratricem* demonstrari *diximus* (*XXX*) tradamus.

Si enim quadratum sit $\alpha\beta\gamma\delta$, et circumferentia $\beta\varepsilon\delta$ circa centrum γ et linea $\beta\eta\vartheta$ quadratrix, ut supra exposuimus, ducta sit, demonstratur esse ut circumferentiam $\beta\varepsilon\delta$ ad rectam $\beta\gamma$, ita rectam $\beta\gamma$ ad $\gamma\vartheta$. Si enim non est, erit *ut circumferentia* $\beta\varepsilon\delta$ *ad rectam* $\beta\gamma$, ita $\beta\gamma$ aut ad maiorem quam $\gamma\vartheta$ aut ad minorem. Prop. 26*)

Sit primum, si fieri possit, ad maiorem, *velut* $\gamma\varkappa$, et circa centrum γ describatur circumferentia $\zeta\eta\varkappa$, quae lineam *quadratricem* in puncto η secet, et *ducatur* perpendicularis $\eta\lambda$, et iuncta $\gamma\eta$ producatur ad ε *punctum* $\beta\varepsilon\delta$ *circumferentiae*. Iam quia *ex hypothesi*, ut $\beta\varepsilon\delta$ circumferentia ad rectam $\beta\gamma$, ita est $\beta\gamma$, id est $\gamma\delta$, ad $\gamma\varkappa$, atque ut $\gamma\delta$ ad $\gamma\varkappa$, ita $\beta\varepsilon\delta$ circumferentia ad $\zeta\eta\varkappa$ circumferentiam — nam circulorum circumferentiae inter se sunt ut diametri [1]) — apparet circumferentiam $\zeta\eta\varkappa$ rectae $\beta\gamma$ aequalem esse. Et quia propter lineae proprietatem (*XXX*) ut circumferentiae $\beta\varepsilon\delta$ $\varepsilon\delta$, ita inter se sunt rectae $\beta\gamma$ $\eta\lambda$, ergo etiam ut circumferentiae $\zeta\eta\varkappa$ $\eta\varkappa$, ita inter se sunt rectae $\beta\gamma$ $\eta\lambda$. Et demonstravimus circumferentiam $\zeta\eta\varkappa$ rectae $\beta\gamma$ aequalem esse; itaque circumferentia $\eta\varkappa$ rectae $\eta\lambda$ aequalis est, quod quidem absurdum. Ergo non est ut $\beta\varepsilon\delta$ circumferentia ad rectam $\beta\gamma$, ita $\beta\gamma$ ad maiorem quam $\gamma\vartheta$.

XXXII. Sed nego etiam ad minorem. Si enim fieri possit, sit ad $\gamma\varkappa$, et circa centrum γ describatur circumferentia

*) Conf. Bretschneider l. c. p. 153 sq.

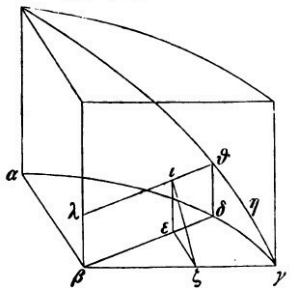
1) Hoc theorema exstat V propos. 11 et VIII propos. 22; simul autem scriptor tacite efficit circulorum arcus quibus aequales anguli insistunt inter se esse ut radios. Vide infra propos. 36 adnot. 1.

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2. *ἀλλὰ Hu*, *πολὺ* ABS (*πολὺ δὲ* voluit *Co*, item *δὲ* post *πρότερον* add. *To*) *παραδοτέον* coni. *Hu* 4. 5. *τοῦ ΑΒΓΔ Co* pro *τοῦ ΑΒΓ* 7. *τῆς δὲ ΒΗΘ To* pro *τῆς δὲ ΒΕΘ* 12. 13. *τῆς ΓΘΗ προσελάσσονα* AB[1], corr. B[2]S 28. *ὡς ἡ ΒΔ* A, sed *E* superscr. prima m. 35. *ΑΒ* A[1] in marg. (S), om. B

γεγράφθω ἡ ZMK, καὶ πρὸς ὀρθὰς τῇ ΓΔ ἡ ΚΗ τέμνουσα τὴν τετραγωνίζουσαν κατὰ τὸ Η, καὶ ἐπιζευχθεῖσα ἡ ΓΗ ἐκβεβλήσθω ἐπὶ τὸ Ε. ὁμοίως δὴ τοῖς προγεγραμμένοις δείξομεν καὶ τὴν ZMK περιφέρειαν τῇ ΒΓ εὐθείᾳ ἴσην, καὶ ὡς τὴν ΒΕΔ περιφέρειαν πρὸς τὴν ΕΔ, τουτέστιν ὡς τὴν ZMK πρὸς τὴν MK, οὕτως τὴν ΒΓ εὐθεῖαν πρὸς τὴν ΗΚ. ἐξ ὧν φανερὸν ὅτι ἴση ἔσται ἡ MK περιφέρεια τῇ ΚΗ εὐθείᾳ, ὅπερ ἄτοπον. οὐκ ἄρα ἔσται ὡς ἡ ΒΕΔ περιφέρεια πρὸς τὴν ΒΓ εὐθεῖαν, οὕτως ἡ ΒΓ πρὸς ἐλάσσονα τῆς ΓΘ. ἐδείχθη δὲ ὅτι οὐδὲ πρὸς μείζονα· πρὸς αὐτὴν ἄρα τὴν ΓΘ.

50 Ἔστι δὲ καὶ τοῦτο φανερὸν ὅτι ἡ τῶν ΘΓ ΓΒ εὐθειῶν τρίτη ἀνάλογον λαμβανομένη εὐθεῖα ἴση ἔσται τῇ ΒΕΔ περιφερείᾳ, καὶ ἡ τετραπλασίων αὐτῆς τῇ τοῦ ὅλου κύκλου περιφερείᾳ. εὑρημένης δὲ τῆς τοῦ κύκλου περιφερείᾳ ἴσης εὐθείας πρόδηλον ὡς δὴ καὶ αὐτῷ τῷ κύκλῳ ῥᾴδιον ἴσον τετράγωνον συστήσασθαι· τὸ γὰρ ὑπὸ τῆς περιμέτρου τοῦ κύκλου καὶ τῆς ἐκ τοῦ κέντρου διπλάσιόν ἐστι τοῦ κύκλου, ὡς Ἀρχιμήδης ἀπέδειξεν.

51 

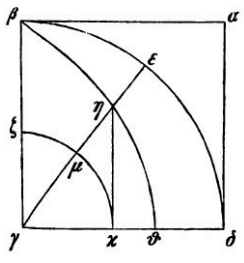
λγ'. Αὕτη μὲν οὖν ἡ γένεσις τῆς γραμμῆς ἐστιν, ὡς εἴρηται, μηχανικωτέρα, γεωμετρικῶς δὲ διὰ τῶν πρὸς ἐπιφανείαις τόπων ἀναλύεσθαι δύναται τὸν τρόπον τοῦτον.

Θέσει κύκλου τεταρτημόριον τὸ ΑΒΓ, καὶ διήχθω, ὡς ἔτυχεν, ἡ ΒΔ, καὶ κάθετος ἐπὶ τὴν ΒΓ ἡ ΕΖ λόγον ἔχουσα

3. δὴ *Hu* pro δὲ 5. 6. τοῦτό ἐστιν AB, corr. S 6. πρὸς τὴν MK οὕτως τὴν ΒΓ εὐθεῖαν bis scripta in A(B), corr. S 13. τρίτη ἀνάλογον λαμβανομένη ABS, corr. *To* 16. δὴ S, δεῖ AB *To* ῥᾴδιον AB, ῥᾳδίως *To* 20 sqq.] cap. 51—56 stilo multis in rebus diverso a vulgari veterum mathematicorum consuetudine composita sunt 20. ΑΓ A^1 in marg. (S), om. B 23. 24. προσεπιφανείαις A (B *To*), πρὸς ἐπιφανείας S (conf. p. 270, 18) 26. τετάρτη μόριον A, coniunx.

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$\zeta\mu\varkappa$, et rectae $\gamma\delta$ perpendicularis *ducatur* $\varkappa\eta$ quadratricem in puncto η secans, et iuncta $\gamma\eta$ producatur ad ε. Similiter



igitur ac supra scripsimus demonstrabimus et circumferentiam $\zeta\mu\varkappa$ rectae $\beta\gamma$ aequalem esse, et ut circumferentias $\beta\varepsilon\delta$ $\varepsilon\delta$, id est $\zeta\mu\varkappa$ $\mu\varkappa$, ita inter se esse rectas $\beta\gamma$ $\eta\varkappa$. Unde apparet circumferentiam $\mu\varkappa$ rectae $\varkappa\eta$ aequalem esse, quod quidem absurdum. Ergo non est ut $\beta\varepsilon\delta$ circumferentia ad rectam $\beta\gamma$, ita $\beta\gamma$ ad minorem quam $\gamma\vartheta$. Nec vero ad maiorem, ut *modo* demonstravimus; ergo ad ipsam $\gamma\vartheta$.

Sed hoc quoque apparet, si rectarum $\vartheta\gamma$ $\gamma\beta$ tertia proportionalis sumatur, hanc circumferentiae $\beta\varepsilon\delta$, itaque quadruplam rectam toti circumferentiae circuli aequalem esse. Rectâ autem, quae circuli circumferentiae aequalis sit, inventâ apparet etiam

ipsi circulo aequale quadratum facile construi posse; Prop. 27
nam rectangulum quod circuli perimetro et radio continetur duplum est circuli, ut Archimedes[2]) demonstravit.

XXXIII. Hic igitur lineae *quadratricis* ortus est, quem Prop. 28*)
magis ad mechanicam rationem accedere diximus (*XXXI*); geometrice autem per locos qui sunt ad superficies[1]) *problema* solvi potest hoc modo.

Sit circuli quadrans $\alpha\beta\gamma$ positione *datus*, et ducatur, ut libet, *a centro ad circumferentiam* recta $\beta\delta$, et perpendicu-

2) Paulo aliis verbis Pappus id theorema enuntiat atque ipse Archimedes circuli dimens. propos. 1: *πᾶς κύκλος ἴσος ἐστὶ τριγώνῳ ὀρθογωνίῳ, οὗ ἡ μὲν ἐκ τοῦ κέντρου ἴση μιᾷ τῶν περὶ τὴν ὀρθήν, ἡ δὲ περίμετρος τῇ λοιπῇ.* Conf. infra V propos. 2 extr. et 3.

*) Conf. Chasles, *Aperçu historique*, p. 28—30 vers. German. Figuram in codicibus corruptam restituit Torellius.

1) Euclidis libros duos hoc titulo inscriptos significari patet ex VII cap. 3.

BS 27. 28. *ωσετύχην* A(B¹), corr. B³S 28. *ἡ* $\overline{B\Delta}$ AB, sed in A Δ vix diversum a Λ, unde *ἡ* $\overline{\beta\lambda}$ S

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```


Reopen these settings in the published workshop

This file opens with scripts disabled and needs no network subresources. The published-workshop link needs the app. Clocks are exact integers; trigonometric values and printed decimals are numerical approximations.

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